

Test 1 Memo

Q1

P.T $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$
a) Let $x \in S \cup (T \cap U)$

$$\Rightarrow x \in S \text{ OR } x \in (T \cap U) \quad \checkmark$$

$$\Rightarrow x \in S \text{ OR } x \in T \text{ and } x \in U$$

$$\Rightarrow x \in S \text{ OR } x \in T \text{ and } x \in S \text{ OR } x \in U$$

$$\Rightarrow x \in (S \cup T) \text{ and } x \in (S \cup U) \quad \checkmark$$

$$\Rightarrow x \in (S \cup T) \cap (S \cup U)$$

$$\Rightarrow S \cup (T \cap U) \subseteq (S \cup T) \cap (S \cup U) \quad \checkmark$$

$$\text{Let } x \in (S \cup T) \cap (S \cup U)$$

$$\Rightarrow x \in (S \cup T) \text{ and } x \in (S \cup U) \quad \checkmark$$

$$\Rightarrow x \in S \text{ OR } x \in T \text{ and } x \in S \text{ OR } x \in U$$

$$\Rightarrow x \in S \text{ OR } (x \in T \text{ and } x \in U) \quad \checkmark$$

$$\Rightarrow x \in S \text{ OR } x \in (T \cap U)$$

$$\Rightarrow x \in (S \cup (T \cap U))$$

$$\Rightarrow (S \cup T) \cap (S \cup U) \subseteq S \cup (T \cap U) \quad \checkmark$$

$$\therefore S \cup (T \cap U) = (S \cup T) \cap (S \cup U) \quad (3)$$

Q1

$$b) S \cap (T \cap U) = (S \cap T) \cap U$$

$$\begin{aligned} S \cap (T \cap U) &= S \cap \{x / x \in T \text{ and } x \in U\} \\ &= \{x / x \in S \text{ and } x \in T \text{ and } x \in U\} \\ &= \{x / x \in S \text{ and } x \in T\} \cap U \\ &= (S \cap T) \cap U \quad \checkmark \end{aligned}$$

OR

$$\begin{aligned} \text{Let } x \in S \cap (T \cap U) &\Rightarrow x \in S \text{ and } x \in (T \cap U) \\ &\Rightarrow x \in S \text{ and } x \in T \text{ and } x \in U \\ &\Rightarrow \text{and } x \in T \quad x \in (S \cap T) \text{ and } x \in U \\ &\Rightarrow x \in (S \cap T) \cap U \\ &\Rightarrow S \cap (T \cap U) \subseteq (S \cap T) \cap U \end{aligned}$$

$$\begin{aligned} \text{Let } x \in (S \cap T) \cap U &\Rightarrow x \in (S \cap T) \text{ and } x \in U \\ &\Rightarrow x \in S \text{ and } x \in T \text{ and } x \in U \\ &\Rightarrow x \in S \text{ and } x \in (T \cap U) \\ &\Rightarrow x \in S \cap (T \cap U) \\ &\Rightarrow (S \cap T) \cap U \subseteq S \cap (T \cap U) \\ &\therefore S \cap (T \cap U) = (S \cap T) \cap U \end{aligned}$$

Q2

For arbitrary $n \in \mathbb{Z}$, $2n+1$ is odd in $\mathbb{Z}$,
and $p(2n+1) = n$. Thus p is onto.

We have $p(2) = p(5) = 2$ ✓
∴ p is not one-to-one. ✓

Since $q(x)$ is always a multiple of 4, there is no $x \in \mathbb{Z}$ such that

~~$q(x) = 3$~~ $q(x) = 3$ and hence q is not onto. ✓

$$q(x) = q(2) \Rightarrow 4x = 4 \cdot 2$$

$$\Rightarrow x = 2$$

∴ q is one-to-one. ✓

$$(p \circ q)(x) = p(q(x)) = p(4x) = 4x$$

This means that ~~$(p \circ q)(x) = q(x)$~~

$$(p \circ q)(x) = q(x) \quad \forall x \in \mathbb{Z}$$

Therefore $p \circ q = q$ is not onto and is one-to-one. ✓

Q2

$$(q \circ p)(x) = q(p(x))$$

$$= \begin{cases} q(x) & \text{if } x \text{ is even} \\ q\left(\frac{x-1}{2}\right) & \text{if } x \text{ is odd} \end{cases}$$

$$= \begin{cases} 4x & \text{if } x \text{ is even} \\ 2(x-1) & \text{if } x \text{ is odd} \end{cases}$$

Since $(q \circ p)(x)$ is never odd,
there is no x such that $(q \circ p)(x) = 1$
and $(q \circ p)$ is not onto. ✓

Also, since $(q \circ p)(2) = (q \circ p)(5) = 8$,
 $(q \circ p)$ is not one-to-one. ✓

Q3

a) For arbitrary $x, y, z \in \mathbb{Z}$

1. $x \equiv x \pmod{4}$, since $x - x = (4)(0)$.
It is Reflexive ✓

2. $x \equiv y \pmod{4} \Rightarrow x - y = 4k$, for some $k \in \mathbb{Z}$
 $\Rightarrow y - x = 4(-k)$ for $-k \in \mathbb{Z}$
 $\Rightarrow y \equiv x \pmod{4}$
It is symmetric ✓

3. $x \equiv y \pmod{4}$ & $y \equiv z \pmod{4}$
 $\Rightarrow x - y = 4k$ and $y - z = 4m$
for some $k, m \in \mathbb{Z}$
 $\Rightarrow x - z = \underline{x - y + y - z} = 4(k + m)$
and $k + m \in \mathbb{Z}$
 $\Rightarrow x \equiv z \pmod{4}$.

It is Transitive & ✓

$\therefore$ Congruence modulo 4 is an equivalence relation.

b) $[0] = \{ \dots -8, -4, 0, 4, 8 \dots \}$ ✓
 $[1] = \{ \dots -7, -3, 1, 5, 9 \dots \}$ ✓
 $[2] = \{ \dots -6, -2, 2, 6, 10 \dots \}$ ✓
 $[3] = \{ \dots -5, -1, 3, 7, 11 \dots \}$ ✓

④

$$x * y = x + y + 3$$

$$y * x = y + x + 3 = x + y + 3$$

$\Rightarrow x * y = y * x \therefore$ It is commutative ✓

$$(x * y) * z = (x + y + 3) * z$$

$$\Rightarrow x + y + 3 + z + 3$$

$$\Rightarrow x + y + z + 6$$

$$x * (y * z) = x * (y + z + 3)$$

$$\Rightarrow x + y + z + 3 + 3$$

$\Rightarrow x + y + z + 6 \therefore *$ is associative ✓

$e = -3$ is the identity element ✓

$$\text{Since } x * -3 = x = -3 * x \checkmark$$

$y = -6 - x$ is the inverse ✓

$$\text{Since } (x) * (-6 - x) = -3 \quad \checkmark$$

$$(-6 - x) * x = -3 \quad \checkmark$$

~~$$x * (-6 - x) = -3$$~~